ECABSD: An Explainable Cross-Attention Framework for Residue-Level Protein-Protein Interaction Binding Site Discovery

Anumala Manigreeva, D. Nayaneesh, Kantam Pavan Sai Reddy, Kura Vignesh Reddy, Vitta Karthikeya

Department of Computer Science and Engineering, Keshav Memorial Institute of Technology

Narayanaguda, Hyderabad, Telangana, India - 500029

Mentor: Mr. Challa Sundeep Babu, Assistant Professor, Department of CSE, KMIT

Abstract—Protein–protein interaction (PPI) binding site prediction remains a central problem in structural bioinformatics because interface residues are not intrinsic to a single protein chain; instead, they depend on the physicochemical and geometric context provided by the binding partner. Most existing predictors either analyze one chain in isolation or fuse pair information only at the final stage, which limits their ability to model partner-specific residue reorganization at the interface. We present ECABSD, an explainable cross-attention framework for residue-level PPI binding site discovery that combines residue graphs, ESM-2 language-model embeddings, GATv2 structural encoding, bidirectional cross-attention, and interpretability modules. Each protein chain is represented as a residue graph with nodes corresponding to amino acids and edges formed when Cα–Cα distances are below 10 Å. Residue features are enriched with ESM-2 embeddings and refined by a six-layer GATv2 encoder with four attention heads, residual connections, GELU activations, and global context pooling. The resulting chain embeddings are then exchanged through bidirectional cross-attention, enabling residues in one protein to be conditioned explicitly on the partner chain before classification. ECABSD is trained using a hybrid objective combining focal loss and soft Dice loss, with AdamW optimization, cosine annealing, warmup scheduling, gradient clipping, and dynamic threshold optimization for imbalanced residue labels. On a dataset of 3,816 protein complexes assembled from PDBBind and a DIPS subset and evaluated on DB5 under random and homology-filtered splits, ECABSD achieves F1 scores of 0.7010 and 0.5797, ROC-AUC values of 0.9373 and 0.8928, and substantial gains over sequence-only and graph-only baselines. Ablation studies show that cross-attention is the most critical component, reducing F1 from 0.7010 to 0.4103 when removed. Explainability analyses using Grad-CAM and attention rollout yield biologically plausible interface maps, supporting the interpretability of learned residue interactions. These results indicate that partner-aware cross-attention is a principled and effective mechanism for residue-level PPI binding site prediction.

Index Terms—protein–protein interaction, binding site prediction, residue-level interface prediction, graph attention network, ESM-2, cross-attention, explainable AI, structural bioinformatics

I. Introduction

Protein–protein interactions regulate signaling, transport, catalysis, immune recognition, and many other cellular processes. Identifying the residues that participate in binding interfaces is therefore fundamental to understanding molecular mechanism and to enabling docking, mutagenesis, and drug discovery. Yet interface prediction is intrinsically difficult because a residue's binding role is relational rather than absolute: the same surface patch can act as an interface in one complex and remain non-interacting in another.

Recent advances in deep learning and protein language models have improved residue representation, but many predictors still treat proteins largely in isolation. Structural graph neural networks can capture local geometry, while transformer-based protein embeddings can encode evolutionary context, but neither alone fully models partner-specific binding logic. This motivates ECABSD, which explicitly exchanges information between the two proteins in a complex through bidirectional cross-attention. The core idea is that a residue should be judged not only by its own local environment, but also by how well that environment complements the partner chain.

This paper makes five contributions. First, it introduces a partner-aware residue-level prediction architecture based on bidirectional cross-attention. Second, it integrates ESM-2 embeddings with a GATv2 structural encoder. Third, it provides an explainability framework using Grad-CAM and attention rollout. Fourth, it reports robust performance under homology-filtered evaluation. Fifth, it outlines an end-to-end deployment design suitable for practical structural biology workflows.

II. Related Work

Deep learning for PPI prediction has progressed from sequence-based CNNs and RNNs toward graph-based and transformer-based models. Broad surveys show that modern methods increasingly rely on structural representations, yet interpretability and generalization remain persistent issues. Graph neural networks are attractive because they can encode residue neighborhoods and 3D proximity, while language models such as ESM-2 provide powerful sequence priors learned from massive protein corpora. Geometric learning benchmarks such as DIPS-Plus have further accelerated the field by supplying large-scale residue-level supervision.

Residue-level interface prediction has traditionally used handcrafted features, solvent accessibility, evolutionary conservation, and local structural descriptors. More recent methods exploit GNNs to propagate information over residue-contact graphs, but many remain single-chain predictors or adopt late fusion of pairwise summaries. That design can miss fine-grained interface coupling, especially for residues whose interaction status depends on the partner geometry. Cross-attention has been successful in multimodal and interaction learning because it allows one entity to query another directly, and this is especially relevant for PPIs where reciprocity is fundamental.

Benchmarking resources also matter. DB5 remains a standard docking and interface benchmark, while DIPS-Plus provides a much larger training resource with residue-level annotations and richer features. These datasets motivate training strategies that are robust to class imbalance and homology leakage. ECABSD is designed specifically for this regime.

III. Research Gap Analysis

Three gaps motivate the present work. First, many residue-level PPI predictors are not truly partner aware. They estimate the likelihood that a residue is "sticky" or surface exposed, but they do not model how binding emerges from complementarity between two proteins. Second, structural encoders often stop at within-chain message passing, which is inadequate for capturing interface formation. Third, interpretability is usually limited to post hoc saliency maps without a clear connection between the prediction pipeline and the biological interaction mechanism.

ECABSD addresses these gaps by explicitly conditioning residue embeddings on the partner chain via bidirectional cross-attention. This is not a cosmetic fusion step; it changes the hypothesis space of the model from "does this residue look like an interface?" to "does this residue become an interface in the context of this partner?" That distinction is essential for PPIs, because molecular recognition is context dependent and often asymmetric at the residue level even if the complex is symmetric at the pair level.

IV. Methodology

A. Problem Formulation

Given a protein complex (P(A), P(B)), ECABSD predicts a binary label $y_i \in \{0, 1\}$ for each residue i in both chains. A residue is labeled as interface if any heavy atom of that residue lies within 4.5 Å of any heavy atom in the partner chain. The model outputs $\hat{p}_i \in [0, 1]$ as the interface probability.

B. Graph Construction

Each chain is modeled as a residue graph G = (V, E), where nodes correspond to residues and edges connect residues whose Cα distance is less than 10 Å. This graph preserves local structural neighborhoods, secondary-structure continuity, and packing interactions while remaining computationally efficient. Node attributes combine sequence and structure-derived features, including ESM-2 embeddings.

C. GATv2 Encoder

A six-layer GATv2 encoder propagates information over the residue graph. Unlike fixed aggregation schemes, GATv2 computes adaptive attention scores conditioned on both source and target node features. Four attention heads enable diverse neighborhood interactions to be learned simultaneously. Residual connections stabilize training, GELU activation improves nonlinear expressiveness, and global context pooling summarizes each chain into a compact chain-level embedding.

D. Cross-Attention Module

Let H(A) and H(B) denote residue embeddings for the two chains. The bidirectional cross-attention module updates chain A using chain B as context and vice versa. This is crucial because interface residues are determined jointly: a residue may be chemically competent for binding, but only the partner chain reveals whether that competence is realized. This module aligns residue neighborhoods across chains, disambiguates ambiguous residues using partner context, and exposes cross-chain interaction structure that can be visualized for interpretability.

E. Explainability Framework

ECABSD combines Grad-CAM and attention rollout. Grad-CAM highlights feature channels that drive the interface score, yielding residue-level saliency maps. Attention rollout aggregates attention across layers, showing which residues exert the greatest downstream influence. Together they provide complementary views of model reasoning, helping verify whether the model focuses on known interface patches, electrostatic complementarity, and conserved hotspot residues.

F. Training Strategy

Training uses a hybrid loss: L = 0.6 L_focal + 0.4 L_dice. Focal loss emphasizes hard, minority interface residues, while soft Dice loss improves overlap on imbalanced labels. Optimization uses AdamW, cosine annealing, linear warmup, and gradient clipping. Dynamic threshold optimization selects the decision threshold that best balances precision and recall on validation data. Chain-swap augmentation improves robustness to protein ordering.

V. Mathematical Formulation

A. Graph Attention

For node i and neighbor j: $e_{ij} = a^T\sigma(W[h_i \parallel h_j])$; $\alpha_{ij} = \exp(e_{ij}) / \Sigma_k \in N(i) \exp(e_{ik})$; $h'_i = \Sigma_j \in N(i)\ \alpha_{ij} W h_j$.

B. Multi-Head Cross-Attention

For query Q, key K, and value V: $Attn(Q, K, V) = softmax(QK^T / \sqrt{dk}) V$. For multi-head attention: $MHA(Q, K, V) = Concat(head_1, \ldots, head_m) W^O$.

C. Loss Functions

Focal loss: $L\_focal = -(1/N) \Sigma_i [\alpha_t(1-\hat{p}_i)^\wedge\gamma\ y_i \log(\hat{p}_i) + (1-\alpha_t)\ \hat{p}_i{}^\wedge\gamma\ (1-y_i) \log(1-\hat{p}_i)]$.

Soft Dice loss: $L\_dice = 1 - [2\ \Sigma_i\ y_i\ \hat{p}_i + \varepsilon] / [\Sigma_i\ y_i + \Sigma_i\ \hat{p}_i + \varepsilon]$.

D. Evaluation Metrics

Precision, recall, F1, MCC, ROC-AUC, and PR-AUC are computed in the standard way. $MCC = (TP\cdot TN - FP\cdot FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$.

VI. Experimental Setup

The dataset contains 3,816 protein complexes assembled from PDBBind and a DIPS subset. Evaluation is performed on DB5 under both random and homology-filtered splits. MMseqs2 filtering at 30% sequence identity reduces train–test leakage and tests whether the model generalizes beyond close homologs. Interface labels are assigned using a 4.5 Å atom-contact cutoff.

The model is trained with a six-layer GATv2 backbone, 256 hidden dimensions, four attention heads, residual connections, global context pooling, and a three-layer MLP classifier with sigmoid output. Performance is measured using accuracy, precision, recall, F1, MCC, ROC-AUC, and PR-AUC.

TABLE I: Main Performance of ECABSD

Split	Acc.	Prec.	Recall	F1	MCC	ROC-AUC	PR-AUC
Random	0.8989	0.6396	0.7756	0.7010	0.6452	0.9373	0.7462
Homology-filtered	0.8828	0.5305	0.6389	0.5797	0.5152	0.8928	0.6077

VII. Results and Discussion

ECABSD achieves strong performance under both evaluation regimes. Under the random split, it reaches 0.8989 accuracy, 0.6396 precision, 0.7756 recall, 0.7010 F1, 0.6452 MCC, 0.9373 ROC-AUC, and 0.7462 PR-AUC. Under the homology-filtered split, it attains 0.8828 accuracy, 0.5305 precision, 0.6389 recall, 0.5797 F1, 0.5152 MCC, 0.8928 ROC-AUC, and 0.6077 PR-AUC.

The drop in performance under homology filtering is expected, but the retained gain over simpler baselines indicates that ECABSD learns meaningful structural interaction patterns rather than relying entirely on sequence similarity. The PR-AUC values are particularly important because interface residues are typically rare and class imbalance makes accuracy alone misleading.

VIII. Component Ablation

This section isolates the contribution of each architectural component relative to the full model; it reflects internal ablations rather than a head-to-head comparison against external published predictors, which is left as future work. Removing cross-attention reduces F1 from 0.7010 to 0.4103, strongly supporting the hypothesis that partner context is essential. Replacing GATv2 with GCN lowers F1 to 0.4891, indicating that adaptive attention-based neighborhood aggregation outperforms static convolution for interface residues. Removing global pooling gives 0.5412 F1, and a sequence-only MLP falls to 0.3847, showing that both structure and partner conditioning are required.

TABLE II: Ablation Study

Model Variant	F1
ECABSD full model	**0.7010**
No cross-attention	0.4103
GCN instead of GATv2	0.4891
No global pooling	0.5412
Sequence-only MLP	0.3847

IX. Explainability Analysis

Explainability results show that ECABSD attends to spatially coherent interface patches rather than diffuse surface regions. Grad-CAM highlights residues contributing most strongly to the final binding score, while attention rollout identifies cross-chain residue pairs with high mutual influence. These heatmaps often coincide with expected interface features such as hydrophobic clusters, electrostatic complementarity, and conserved contact hotspots.

X. Illustrative Example

For complex 1AY7, ECABSD reaches precision 0.938, recall 1.000, and F1 0.968. This single-complex example illustrates the kind of residue-level localization the model can achieve; broader validation across a larger case set is needed before generalizing this result. The residue maps produced by the explanation modules localize the predicted interface to the expected contact region.

XI. Deployment Architecture

ECABSD is designed for deployment through a FastAPI service connected to docking workflows. The user uploads a protein complex, the system builds residue graphs, computes embeddings, predicts interface probabilities, and returns annotated residues and cross-chain heatmaps.

XII. Limitations

The model still depends on structural inputs or reliable structure-derived graphs, which limits applicability to highly disordered proteins or incomplete complexes. Although homology filtering reduces leakage, residual dataset bias may remain. ECABSD predicts residue-level binding propensity rather than explicit atom-level contact geometry, so it complements rather than replaces docking methods. No external baseline comparison or cross-fold variance is reported in the current evaluation.

XIII. Future Work

Future extensions should incorporate equivariant transformers, uncertainty calibration, and multimeric context beyond pairwise complexes, along with a direct comparison against published predictors such as DELPHI and MaSIF-site, and reporting of cross-fold variance.

XIV. Conclusion

ECABSD provides a partner-aware and interpretable framework for residue-level PPI binding site discovery. By integrating ESM-2 embeddings, GATv2 structural encoding, bidirectional cross-attention, and saliency-based explanation methods, it captures both local structural context and inter-chain dependence. Its performance under random and homology-filtered evaluation, together with ablation evidence, supports the claim that explicit partner-aware cross-attention is a principled mechanism for interface prediction.

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